unchanging straight-line trajectory —*constant velocity motion*, it's called —you can feel *changes* to your velocity.

But if you think about it for a moment, there is something odd about this. What is it about changes in velocity that allows them to stand alone, to have intrinsic meaning? If velocity is something that makes sense only by comparisons—by saying that *this* is moving with respect to *that*—how is it that changes in velocity are somehow different, and don't also require comparisons to give them meaning? In fact, could it be that they actually *do* require a comparison to be made? Could it be that there is some implicit or hidden comparison that is actually at work every time we refer to or experience accelerated motion? This is a central question we're heading toward because, perhaps surprisingly, it touches on the deepest issues surrounding the meaning of space and time.

Galileo's insights about motion, most notably his assertion that the earth itself moves, brought upon him the wrath of the Inquisition. A more cautious Descartes, in his *Principia Philosophiae*, sought to avoid a similar fate and couched his understanding of motion in an equivocating framework that could not stand up to the close scrutiny Newton gave it some thirty years later. Descartes spoke about objects' having a resistance to changes to their state of motion: something that is motionless will stay motionless unless someone or something forces it to move; something that is moving in a straight line at constant speed will maintain that motion until someone or something forces it to change. But what, Newton asked, do these notions of "motionless" or "straight line at constant speed" really mean? Motionless or constant speed with respect to what? Motionless or constant speed from whose viewpoint? If velocity is not constant, with respect to what or from whose viewpoint is it not constant? Descartes correctly teased out aspects of motion's meaning, but Newton realized that he left key questions unanswered.

Newton—a man so driven by the pursuit of truth that he once shoved a blunt needle between his eye and the socket bone to study ocular anatomy and, later in life as Master of the Mint, meted out the harshest of punishments to counterfeiters, sending more than a hundred to the gallows—had no tolerance for false or incomplete reasoning. So he decided to set the record straight. This led him to introduce the bucket.[1]